

[1] Find errors, if any, in the following declaration statements.

```
Int x;  
float Letter, DIGIT;  
double = p,q  
exponent alpha , beta;  
m, n, z:INTEGER  
short char c;  
long int m; count;  
long float temp;
```

Solution:

```
int x; // 'Int' should be 'int' (C is case-sensitive).  
float Letter, DIGIT; // Correct as is.  
  
double p, q; // Remove '='. Variables should be separated by ','.  
  
double exponent, alpha, beta; // 'exponent' is not a valid type.  
Use 'double' instead.  
  
int m, n, z; // 'INTEGER' is not a valid type in C. Use 'int'.  
  
short char c; // 'short char' is an invalid combination. Use  
'short c;' or 'char c;'.  
  
long int m, count; // Fix declaration: 'count' should be  
included in the type.  
  
float temp; // 'long float' is incorrect. Use 'float' because 'long  
float' does not exist.
```

[2] Identify syntax errors in the given program.
After corrections, what output would you expect
when executed?

```
#define PI 3.14159
main()
{
    int R, C; /* R - Radius of Circle */
    float perimeter; /* Circumference of Circle */
    float area; /* Area of Circle */

    C = PI;
    R = 5;
    Perimeter = 2.0 * C * R;
    Area = C * R * R;

    printf("%f", "%d", &perimeter, &area);
}
```

Solution:

- Fixed declaration errors (Int → int, long float → float, etc.).
- Corrected variable names (Perimeter → perimeter, Area → area).
- Fixed printf() statement (used correct format specifiers).
- Ensured proper function return type (int main()).

Corrected Code:

```
#include <stdio.h>
#define PI 3.14159

void main()
{
    int R;
    float C, perimeter, area;
    /* R - Radius of Circle, C - PI value */

    C = PI;
    R = 5;

    perimeter = 2.0 * C * R;
    area = C * R * R;

    printf("Perimeter = %f, Area = %f", perimeter, area);
}
```